

AARON A

Full Stack Developer | MERN Stack Developer | React Developer | Microservices & Multi-Agent AI

Chennai, Tamil Nadu | +91 6382315385 | aaronbca123@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | LeetCode (50+ solved) | HackerRank (30+ solved)

PROFESSIONAL SUMMARY

Full Stack Developer / MERN Stack Developer with hands-on experience across two full-stack internships, delivering production-grade web applications, RESTful APIs, and microservices. Proficient in React.js, Next.js, Node.js, Express.js, MongoDB, MySQL, TypeScript, and Redux Toolkit, with practical exposure to Supabase (Postgres, Row-Level Security), JWT authentication, Role-Based Access Control (RBAC), the Saga pattern, Circuit Breaker pattern, event-driven architecture (Apache Kafka, RabbitMQ), and autonomous multi-agent AI workflows using LangGraph. Completed a structured 12-week / 480-hour full stack training program spanning frontend, backend, database, DevOps, and business intelligence domains, culminating in a capstone application with a Domo BI dashboard, Docker containerization, and CI/CD deployment. Solved 50+ problems on LeetCode and 30+ on HackerRank (including the Blind 75 list), with strong fundamentals in Data Structures & Algorithms, OOP, DBMS, Operating Systems, and Computer Networks.

TECHNICAL SKILLS

- **Languages:** JavaScript (ES6+), TypeScript, Python, HTML5, CSS3, SQL
- **Frontend:** React.js, Next.js (App Router, SSR), Redux Toolkit, Context API, React Router, React Hooks, Tailwind CSS, Bootstrap, Framer Motion, Responsive/Mobile-First Design, Cross-Browser Compatibility
- **Backend & Architecture:** Node.js, Express.js, REST API Design, MVC Architecture, Microservices, Saga Pattern (Orchestration & Choreography), Circuit Breaker Pattern, Event-Driven Architecture, WebSockets, Webhooks, Middleware Design
- **Messaging & Protocols:** Apache Kafka, RabbitMQ, AMQP, Pub/Sub, TCP, HTTP/HTTPS
- **Databases:** MongoDB, Mongoose ODM, MySQL, Sequelize ORM, Supabase (Postgres, Row-Level Security), Aggregation Pipelines, Database Design & Normalization, Query Optimization & Indexing
- **AI & Agentic Workflows:** Multi-Agent Systems, LangGraph, LLM Agent Integration, Document/PDF Parsing, Twilio SMS API, Nodemailer
- **Business Intelligence:** Domo Platform (Workflows, Dashboards, AppDB, Beast Mode), Data Visualization, KPI Tracking, ETL Processes
- **DevOps & Tools:** Git, GitHub (branching, PRs, code review), Docker, Docker Compose, Postman/Newman, CI/CD (GitHub Actions), Render, VS Code, Chrome DevTools
- **Testing & Methodology:** Jest, Unit Testing, API Testing, Agile, Scrum, Sprint Planning, Code Reviews
- **CS Fundamentals:** OOP, Data Structures & Algorithms (Blind 75), DBMS, Operating Systems, Computer Networks, Big-O Complexity

PROFESSIONAL EXPERIENCE

Software Engineer Trainee | GWC Data AI, ELCOT IT Park, Hosur, Tamil Nadu

Jul 2026 – Oct 2026

- Completed a structured 12-week (480-hour) full stack training program: HTML5/CSS3 and responsive design, React.js (Hooks, component patterns, rendering optimization), and Tailwind CSS.
- Practiced Git branching/merging/rebasing, GitHub collaboration (pull requests, code reviews, issue tracking), and built Postman collections for API testing and automation.
- Built a full stack application integrating Supabase (Postgres backend) with Row-Level Security (RLS) policies, authentication, and file storage, plus a separate production-style app with JWT auth, database integration, and WebSocket-based real-time chat.
- Built RESTful APIs with Node.js/Express.js (routing, middleware, error handling, async patterns) and implemented JWT authentication with refresh tokens, bcrypt password hashing, custom middleware, and Role-Based Access Control (RBAC).
- Designed schemas and wrote complex queries, joins, and transactions across MySQL (Sequelize ORM) and MongoDB (Mongoose ODM), including aggregation pipelines and indexing for performance.
- Developed a TypeScript React application with Redux Toolkit (slices, async thunks, middleware) for type-safe state management and API integration.
- Built an interactive Domo BI dashboard consuming a custom REST API, with multiple visualization types, automated alerts, and an executive summary report.
- Practiced advanced SQL (CTEs, window functions, subqueries, normalization) and query optimization/indexing strategies; documented performance benchmarks.
- Containerized applications with Docker/Docker Compose, built CI/CD pipelines with GitHub Actions, and applied production security practices (HTTPS, CORS, rate limiting, Helmet.js, input validation).
- Delivered a capstone project — a full stack app (React + TypeScript + Redux, Node.js/Express, MySQL/MongoDB) with a Domo BI dashboard, Docker containerization, and CI/CD deployment — and presented the technical architecture in a code review defense.

MERN Stack Developer Intern | WHY Global Services, Chennai, India

Jan 2026 – Jul 2026

- Architected and deployed 5+ production-grade RESTful APIs (Node.js, Express.js, MongoDB) covering user authentication, product catalogs, payment processing, and order tracking, verified via Postman.
- Engineered 5–10 reusable React.js/TypeScript components using React Hooks, Redux Toolkit, and Context API, following MVC architecture and OOP principles, reducing repeated UI work across the team.
- Implemented JWT Authentication with Role-Based Access Control (RBAC) to secure all API endpoints and enforce user-level authorization policies.

- Improved responsive design and cross-browser compatibility across mobile, tablet, and desktop using Tailwind CSS for a consistent, accessible user experience.
- Collaborated in full Agile/Scrum workflow (sprint planning, daily stand-ups, peer code reviews) using Docker for containerized development and Git/GitHub for version control.
- Authored Jest unit tests to validate API response integrity and edge cases; integrated Nodemailer for transactional email notifications; deployed on Render following CI/CD best practices.
- Optimized MongoDB aggregation pipelines for admin analytics, inventory management, and sales reporting dashboards, reducing data retrieval complexity.

FEATURED PROJECTS

- **Autonomous Multi-Agent Inventory & Stock Management System** (*Node.js, Express, Sequelize, MySQL, LangGraph, Twilio API, Nodemailer*) — Architected an event-driven replenishment system with autonomous AI agents that monitor stock thresholds, predict depletion velocity, and auto-dispatch Purchase Orders via SMS/email; built a document-parsing agent evaluating 10+ supplier PDFs simultaneously and a Human-in-the-Loop approval gate.
- **Automated Insurance Claim Verification & Analytics Dashboard** (*Domo Workflows, Computer Vision/OCR Agents, AppDB*) — Engineered an automated claim verification pipeline reducing manual audit turnaround from 3 days to near real-time; integrated AI image-analysis agents to flag discrepancies and built interactive Domo dashboards for claim throughput and fraud alerts.
- **Full Stack E-Commerce Website** (*React.js, Redux Toolkit, Node.js, Express.js, MongoDB, JWT, RBAC, Nodemailer, Tailwind CSS*) — Designed and deployed a scalable e-commerce platform with cart management, payment gateway integration, and real-time order tracking; automated transactional emails and deployed with a CI/CD pipeline on Render.
- **Real-Time Chat Application** (*Next.js 14 (App Router), WebSocket, Node.js, MongoDB, JWT, Tailwind CSS, SSR*) — Built a scalable real-time chat application with instant bi-directional messaging, JWT-authenticated route protection, and a responsive SSR-rendered UI.
- **Enterprise Office Task Management Platform** (*React.js, Tailwind CSS, Node.js, Express.js, REST APIs, RBAC*) — Developed an internal portal for hierarchical workflow tracking and multi-tier task delegation with secure RBAC and CRUD endpoints for deadlines, priorities, and progress states.
- **Rytail — Multi-Cycle Retail Agentic System UI** (*JavaScript, Excel data pipelines*) — Building a UI that generates cycle-specific single-file HTML applications from structured Excel data, using a shared app-shell pattern across cycles.
- **Aura** (*React 19, Vite, Tailwind v4, LocalStorage*) — Built a full-featured glassmorphism productivity app for portfolio purposes with local persistence.
- **Strata** (*React, Vite, Tailwind v3*) — SaaS landing page demonstrating Context API, custom hooks, higher-order components, render props, and compound components.
- **Event-Driven Architecture Demo** (*Node.js, React/Vite*) — Built a learning project covering the EventEmitter/publisher-listener pattern for event-driven communication.
- **Interactive Task Manager & Daily Notes SPA** (*React.js, Tailwind CSS, Browser LocalStorage*) — Created a zero-latency, serverless daily task/notes SPA with modular components and dark/light themes, fully usable offline.
- **Developer Portfolio & Landing Page** (*React.js, Tailwind CSS, Web Animations*) — Built an interactive, animated personal brand site optimized for performance and responsiveness across devices.
- **Snake Game** (*React.js, React Hooks, CSS3*) — Engineered a fully functional Snake game demonstrating component-based architecture, keyboard event handling, and collision detection.

PROBLEM SOLVING & COMPETITIVE PROGRAMMING

- LeetCode: Solved 50+ problems across Arrays, Two Pointers, Strings, HashMaps, Trees, Graphs, and Dynamic Programming; practicing the Blind 75 list.
- HackerRank: Solved 30+ challenges in algorithmic optimization, SQL query optimization, and core programming paradigms.
- Primary practice language Python, with an active goal of transitioning problem-solving practice to JavaScript.

CERTIFICATIONS

- Full Stack Development Certification — Whytap Institute

EDUCATION

Bachelor of Computer Applications (BCA) — 80%

Raja College of Arts and Science, Ramanathapuram, Tamil Nadu (Alagappa University)

2025